

Question

M is a metallic element that can form a salt $\text{MX}_2 \cdot n\text{H}_2\text{O}$ (n is an integer) with a monobasic organic acid **HX**. The following thermogravimetric analysis experiment was performed on the salt (all data are experimental values):

(1) $\text{MX}_2 \cdot n\text{H}_2\text{O}$ is heated to 180 °C in nitrogen, with a weight loss of 14.2%, corresponding to the loss of all crystal water; heating is continued to 280 °C, with a further weight loss of 13.4% (Reaction 1); heating is continued to above 400 °C, and after cooling, the remaining solid is metal **M**, with a mass of 54.8% of the original mass.

(2) $\text{MX}_2 \cdot n\text{H}_2\text{O}$ is heated to above 400 °C in air, and after cooling, the remaining solid is oxide **Y**, with a mass of 58.7% of the original mass.

Through calculation, deduce what metal element **M** is, and deduce what substances the salt $\text{MX}_2 \cdot n\text{H}_2\text{O}$ and **Y** are. (Hint: The ratio of the number of metal atoms to oxygen atoms in **Y** is a simple integer ratio)

Which of the following statements are correct:

1. If **M** is in the a -th period and the b -th column from left to right, then $(a+b)/ab > 0.25$
2. MX_2 is a strong electrolyte
3. **Y** is insoluble in NaOH
4. The sum of the number of all atoms in the chemical formula of the solid product of Reaction 1 is less than or equal to 15

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 1,2

G. 1,3

H. 1,4

I. 2,3

J. 2,4

K. 3,4

L. 1,2,3

M. 1,2,4

N. 1,3,4

O. 2,3,4

P. 1,2,3,4

Answer

Correct Answer: **A**

Detailed Explanation

Let the molar mass of **M** be x g/mol, then: $18.02n : x = 14.2 : 54.8$

When $n = 1$, $x = 69.5$ is Ga (oxidation state does not match, discard)

When $n = 2$, $x = 139$ is La (oxidation state does not match, discard) or Ba

When $n = 3$, $x = 208.6$ is Bi (oxidation state does not match, discard) or Pb

Combined with oxide data, let the chemical formula of oxide **Y** be MO_y , then:

$$(x + 16.00y) : x = 58.7 : 54.8$$

When **M** is Ba, solving for $y = 0.60$, which is unreasonable, discard;

When **M** is Pb, solving for $y = 0.92 \approx 1$, corresponding to PbO ;

Therefore, metal **M** is Pb, **Y** is PbO

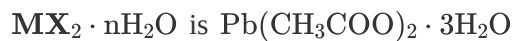
CHECKPOINT

2 PTS

M is Pb, **Y** is PbO

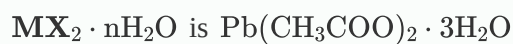
The molar mass of the acid radical ion is: $(207.2 \div 0.548 - 207.2 - 3 \times 18.02)/2 = 58.4$ (g/mol)

Organic acids are often carboxylic acids, and the molecular weight of —COO^- is 44; therefore, 58.4 corresponds to acetate, and HX is acetic acid



CHECKPOINT

1 PTS



$a=6$, $b=14$, $(a+b)/ab=0.238$, statement 1 is incorrect

CHECKPOINT

0.5 PTS

$$(a+b)/ab=0.238$$

$\text{Pb}(\text{CH}_3\text{COO})_2$ is a weak electrolyte, statement 2 is incorrect

CHECKPOINT

0.5 PTS

$\text{Pb}(\text{CH}_3\text{COO})_2$ is a weak electrolyte

PbO is amphoteric and soluble in NaOH , statement 3 is incorrect

CHECKPOINT

0.5 PTS

PbO is amphoteric and soluble in NaOH

The molar mass of the solid product obtained after weight loss at 280 °C in nitrogen is (assuming it contains 1 Pb):

$$(207.2 + 2 \times 59.04 + 3 \times 18.02) \times (1 - 0.142 - 0.134) = 274.6 \text{ (g/mol)}$$

This molar mass corresponds to 1 Pb atom, 1 acetate ion, and 0.5 oxygen atoms. Therefore, the solid product of reaction 1 is $\text{Pb}_2\text{O}(\text{CH}_3\text{COO})_2$, and the sum of the number of all atoms in the chemical formula is 17, statement 4 is incorrect

CHECKPOINT

0.5 PTS

The solid product of reaction 1 is $\text{Pb}_2\text{O}(\text{CH}_3\text{COO})_2$

Choose A